THE UNITED REPUBLIC OF TANZANIA
THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
NAMANGA FORM THREE PRE-MIDTERM EXAMINATION

032 CHEMISTRY 1

Time: 3 Hours **August 2024**

INSTRUCTIONS

1. This paper consists of sections A, B, and C with total **eleven (11)** questions.
2. Answer all questions in section A and B and **two (2)** questions from section C.
3. Section A carry **Sixteen(16)** marks, Section B carry **fifty four (54)** marks and Section C carry **thirty (30)** marks
4. Cellular phones and any unauthorized materials are not allowed in examination room.
5. Write your **examination number** on every page of your answer sheet(s)
6. The following constants may be used:-
   Atomic mass: H=1, C=12, N=14, O=16, Na=23, Ca = 40, S=32, Mg=24, Cl=35.5, K =, Pb = 207, Fe = 56
   Avogadro's number = $6.02 \times 10^{23,}$ GMV at S.T.P = $22.4dm^3$, 1Faraday = 96,500coulombs.
   Standard pressure = 760mm Hg, Standard temperature = 273K, 1 litre = $1dm^3 = 1000cm^3$.

SECTION A (16 marks)

1. For each of the items (i) – (x), choose the correct answer from among the given alternatives and write its letter besides the item number in the answer sheet provided

i. The following are the uses of chromatography **except**
   A. To analyse blood in crime scenes.
   B. To detect different fibres.
   C. To detect water pollution.
   D. To bleach dye/colour.
   E. To test the purity of organic substances.

ii. Which one of the following is the characteristic of covalent bonding?
   A. Electrons are transferred from one atom to another
   B. Electrons are shared by different atoms
   C. Protons are transferred from one atom to another
   D. Protons are shared by different atoms
   E. Ions are formed

iii. The process of adding chemicals in water to dump suspended particles so they can fall and coagulate is;
   A. Aeration B. Stagnation C. Flocculation D. Chlorination
   E. Froth Floatation

iv. In an attempt to assess your learning of practical lessons, your teacher asked you to prepare Oxygen in the laboratory while watching you. Given the following sets of apparatus and chemicals, which one would you use?

A. Bunsen burner, test tube, Potassium chlorate, Manganese (IV) oxide gas jar, water trough with water, and delivery tube

B. Manganese (IV) oxide, zinc granules, dilute sulphuric acid, gas jar, water trough with water, and delivery tube

C. Bunsen burner, zinc granules, dilute hydrochloric acid, gas jar, water trough with water, and delivery tube

D. Thistle funnels, zinc granules, dilute hydrochloric acid,, gas jar, water trough with water, and delivery tube

E. Iron filings, dilute hydrochloric acid, gas jar, water trough with water, and delivery tube

v. Why does it become more difficult to remove electrons from an element on moving to the right across the periodic table?

A. Because the number of neutrons and electrons increase

B. Because the number of electron shells increases

C. Because the number of electron shells decreases

D. Because the number of protons and electrons increase

E. Because the number of protons and electrons decrease.

vi. Which of the following is correct about sub-atomic particles?

A. Electrons have no charge and have negligible weight, protons are positively charged
B. Protons are positively charged, electrons are negatively charged and have significant weight
C. Electrons are negatively charged and have negligible weight, neutrons are positively charged
D. Protons are negatively charged, electrons are positively charged and have negligible weight
E. Electrons are negatively charged and have negligible weight, protons are positively charged

vii. When a certain amount of sodium chloride was dissolves in water, total of 1.2×10^{24} ions of sodium and chloride were formed. What was the mass of the sodium chloride?

A. 58.5 grams
B. 56.5 grams
C. 21 grams
D. 35.5 grams
E. 1.2 grams

viii. When a glass of water is allowed to stand for some time in a warm room, gas bubbles accumulate on the inside surface of a glass. This suggests that:

A. Water molecules decompose to give oxygen gas after absorbing heat.
B. Water molecules are found in air
C. Gases have a greater affinity for glass than liquid water.
D. Liquid water has a greater affinity for glass than water.
E. Gases in water are less soluble in hot water than cold water

ix. Why oxygen differs from other gases?

A. It neither burn nor support combustion
B. It supports combustion but does not burn
C. It burns but does not support combustion
D. It burns and supports combustion
E. It is colourless

x. The process of absorbing moisture from the atmosphere to form a solution is called…..

A. Vaporization
B. Deliquescence
C. Hygroscopic
D. Sublimation
E. Efflorescence

2. Match the items in LIST A with the corresponding items in LIST B by writing the letter of correct response besides the item number in the answer sheet provided.

List A	**List B**
(i) Changes from solid directly to gas when heated (ii) A change in state of a substance from gas (vapour) to liquid (iii) The reverse of sublimation. (iii) An alloy formed by <u>iron</u> (metal) and <u>carbon</u> (non-metal). (iv) A cloudy mixture of tiny droplets of one liquid suspended in another liquid (v) A change in state from solid to liquid	A. Melting B. Ammonium Chloride C. Deposition D. Suspension E. Emulsion F. Condensation G. Brass H. Steel I. Ice

SECTION B (54 marks)

3. (a) Whenever visiting Chemistry laboratory, Seypalm noticed the doors always opened outwards there was a fume chamber inside. Briefly explain the reasons for the doors working that way and the presence of the observed structure.

(b) During the visits to the Chemistry laboratory, Seypalm was also being instructed on the important laboratory safety measures to obey. State the measures (give six points).

4. (a) Briefly explain two amendments made on Daltons atomic theory

(b) Boron has two naturally occurring isotopes: ^{10}B with an atomic mass of 10.01 amu and the percentage abundance of 20%, and ^{11}B with an atomic mass of 11.01 amu and the percentage abundance of 80. If you were asked to construct a periodic table, what atomic mass would you show for Boron on that periodic table?

5. (a) (i) Differentiate between chemical reactions and chemical equations.

(ii) When carbon dioxide was dissolved in an aqueous solution of lime water, the mixture reacts to yield a milky solution. State what causes the formation of the milky solution and write balanced molecular equation, complete ionic equation, and net ionic equations for this process.

(b) State the types of each of the following chemical reactions:

(i) $Fe_{(s)} + S_{(l)} \rightarrow FeS_{(s)}$

(ii) $CaCO_{3(s)} \rightarrow CaO_{(s)} + CO_{2(g)}$

(iii) $CaCl_{2(aq)} + AgNO_{3(aq)} \rightarrow Ca(NO_3)_{2(aq)} + AgCl_{(s)}$

(iv) $PbO_{(s)} + H_{2\,(g)} \rightarrow Pb_{\,(s)} + H_2O_{(l)}$

6. Study the following parts of the periodic table and then answer the questions that follows:
Note: The letters used are not scientific symbols for the elements concerned.

Group I	II	III	IV	V	VI	VII	O
						N	
	K				Q		P
L							

(a) Identify and write the electronic configuration for the elements K, N, P and L.
(b) What type of bond will exist in a compound formed when Q combines with L? Write the chemical formula for the compound formed and list two chemical properties of the compound.

7. (a) Define hard water.
(b) You are visiting village **A** and village **B**, with shortage of potable water supply, and therefore residents have to use brackish water (water naturally containing salts) from their wells for washing clothes; this leads to use of too much soap and to the spoiling of their clothes. When the residents of village **A** boil their brackish water before using them for washing their clothes, the boiled water behaves like fresh water after cooling; , however, when the residents of village **B** try to boil the water before using it for washing their clothes, the water still remains brackish.

(i) With the aid of a chemical equation, briefly explain to the residents of village **B** why boiling can change the brackish water from village **A** to fresh water, but cannot change the brackish water from their wells into fresh water.

(ii) With the aid of chemical equations, briefly explain to the residents of village **B**, method that involves the use of chemicals, they can use to change the brackish water from their wells into fresh water.

8. (a) (i) State the Avogadro's law.
(ii) Convert 5.6 dm^3 of ammonia gas to the actual number of ammonia (NH_3) molecules,
(b) Thermal decomposition of calcium carbonate can be represented by the following equation:
$CaCO_{3(s)} \longrightarrow CaO_{(s)} + CO_{2(g)}$
Calculate the mass left at the end of reaction when 150g of calcium carbonate were completely decomposed.

SECTION C (30 marks)

9. (a) Give the meaning of emulsion, alloy, distillate and sublimation, and then explain four differences between solutions and suspensions.
(b) Kinetic theory of matter is one of the most popular and fundamental theories. Outline its three main points

10. Draw a labelled diagram illustrating the laboratory production of hydrogen gas using an acid and zinc metal, and outline any four properties and five uses of hydrogen.

11. Change in the states of matter is one of the most important phenomenon that makes life possible; explain in six points.